Sagnik Pal

MSc in Computer Science, RKMVERI, Belur

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LinkedIn
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SUMMARY

Motivated and technically driven student with a strong foundation in machine learning and deep learning, along with hands-on experience in developing end-to-end AI and data-driven solutions. Skilled in problem-solving, model development, training, and evaluation, with a proactive, collaborative, and professional approach to real-world technical challenges.

EDUCATION

Ramakrishna Mission Vidyamandira, Belur Math
BSc (Honours) in Computer Sceience, 2022–25
CGPA – 9.45

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Computer Science, 2025–27
CGPA – 7.64(till Sem 2)

SKILLS

- **Languages:** Python, JavaScript, C, C++, Java, SQL, LaTeX, php, Assembly Language
- **Frameworks:** Scikit-Learn, NumPy, Pandas, Streamlit, PyTorch, Matplotlib, Scipy, Django, LangChain, MCP, FastAPI, CrewAI, NetworkX, OpenCV
- **Operating Systems:** Windows, Linux
- **Tools:** Git/GitHub, MS Office, Redis, Apache XAMPP, Linux Command Line, neo4j

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, Emerging Technologies : LLMs, Block Chain, Mining of Massive Datasets
- Probability and Statistics, Linear Algebra, Discrete Mathematics, Algorithms, Compiler Design
- Operating System, Networking, Database Management System, Software Engineering, Data Structures

ACHIEVEMENTS

- CUET-PG 2025 (Computer Science) – Qualified
- Secured Top 5 percentile in NPTEL course Programming In Java
- Presented "Beyond the Hub: Exploiting Structural Bridges for Targeted Epidemic Control via Graph Neural Network" at the Ajit Diwan Memorial Workshop (2026).

Certificates

- NPTEL – Programming In Java (2024)
- NPTEL – Data Base Management System (2023)
- Forage – Deloitte Australia - Technology Job Simulation (2026)

INTERSHIP

- Summer Intern at WEJOY Innovation Ventures Pvt. Ltd.
- Intern at Centre of Excellence on Data Science and Machine Learning, WEBEL, Government of West Bengal

PROJECTS

Ankyra: PDF-Based Dynamic Graph RAG

RKMVERI September 2026 – October 2026

A PDF-based question-answering system that transforms documents into document-specific hybrid ontologies and knowledge graphs using NLP, entity-relation extraction, and attribute extraction. Implemented knowledge graph-based retrieval using NetworkX and RapidFuzz, integrated with NVIDIA Nemotron for generating context-aware, evidence-grounded natural-language answers.

Stock-LLM: Stock Market Forecasting and LLM-based Decision Support using iTrans former and Temporal Fusion Transformer (TFT)

RKMVERI Feb 2026 – May 2026

The Stock-LLM project addresses this challenge by combining deep learning-based time-series forecasting with a large language model (LLM) driven explanation layer. The project demonstrates how time-series forecasting, ensemble learning, sentiment analysis, and LLM based reasoning can be combined into one end-to-end intelligent market analytics system

NeuroGraph-X: A Hybrid CNN-GNN Framework for pMCI vs. sMCI Classification with Natural Language Explainability

RKMVERI Feb 2026 – Apr 2026

This project proposes NeuroGraph-X, a hybrid deep learning framework that integrates 3D Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), and Natural Language Processing (NLP) for accurate and explainable AD detection from structural MRI scans. This approach aims to bridge the gap between predictive performance and medical interpretability

NetGuardAI: Containing Information Contagions A Comparative Study of Heuristic, Planning, and Deep Reinforcement Learning Strategies

RKMVERI Sep 2025 – Dec 2025

NetGuard AI is a research-oriented Reinforcement Learning framework designed to optimize the containment of misinformation (rumors) on complex social networks. By simulating SIR dynamics on modular scale-free networks, our research uncovered the "Hub Fallacy"—demonstrating that while degree-based heuristics target influencers, our GraphSAGE Agent achieves superior containment by identifying and blocking "bridge" nodes that connect isolated communities.

FaceTrack

August 2025 – October 2025

Full-stack Django application leveraging DeepFace for asynchronous video-based face authentication and multi-segment person search. The system processes uploaded videos in the background, retrieves and matches facial embeddings for identity verification of individuals across video segments.